MOTOR HEALTH MONITORING SYSTEM

Using Fuzzy Logic & Genetic Programming

Group Members

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INDUCTION MOTOR

SMART MONITORING

Electrical Engineering | PBL Project
Semester – 5

2. Background & Problem Statement

BACKGROUND

Induction motors are widely used in industries, plants and daily life applications. Unexpected motor failures can lead to production loss, high maintenance cost and safety risks.

Traditional monitoring methods are often rule based and may not adapt well to changing operating conditions.

Fuzzy logic provides intelligent decision making with uncertain information, while genetic programming can optimize the rule set.

PROBLEM

Design an intelligent Motor Health Monitoring System that takes motor current, temperature and vibration as inputs and classifies the motor condition into Normal, Warning or Critical using Fuzzy Logic and optimizes the rule base using Genetic Programming.

OUTPUT

Normal • Warning • Critical

3. Objectives

01 Identify key health parameters of an induction motor.

02 Design a fuzzy logic system for motor health evaluation.

03 Implement and simulate the system in Python.

04 Optimize the fuzzy rule base using genetic programming.

05 Develop a simple interface to show motor health status.

4. Learning Outcomes

01 Understand fuzzy logic and its applications.

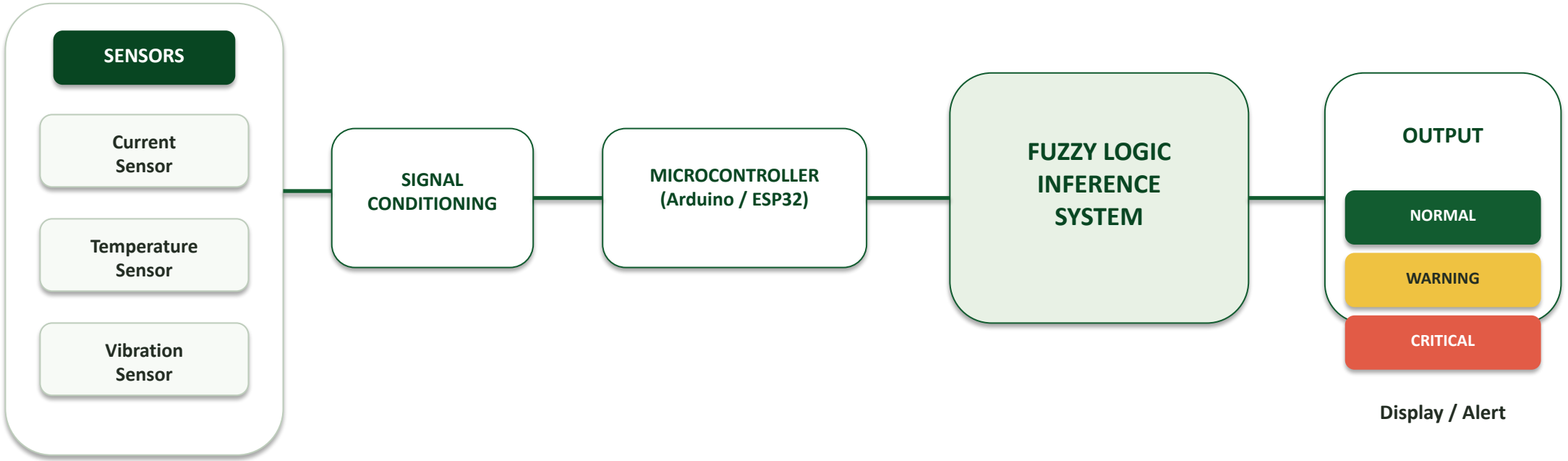
02 Design membership functions and fuzzy rule base.

03 Implement inference system in Python.

04 Apply genetic programming for rule optimization.

05 Analyze results and communicate through reports and presentation.

5. System Design / Block Diagram



6. Input & Output Variables

INPUT VARIABLES

Parameter	Range	Unit
Current (I)	0 – 20	Ampere
Temperature (T)	25 – 100	°C
Vibration (V)	0 – 10	mm/s

OUTPUT VARIABLE

Output	Description
Normal	Motor is healthy and safe to operate
Warning	Motor has warning; needs attention
Critical	Motor is in critical condition; stop required

INTERPRETATION

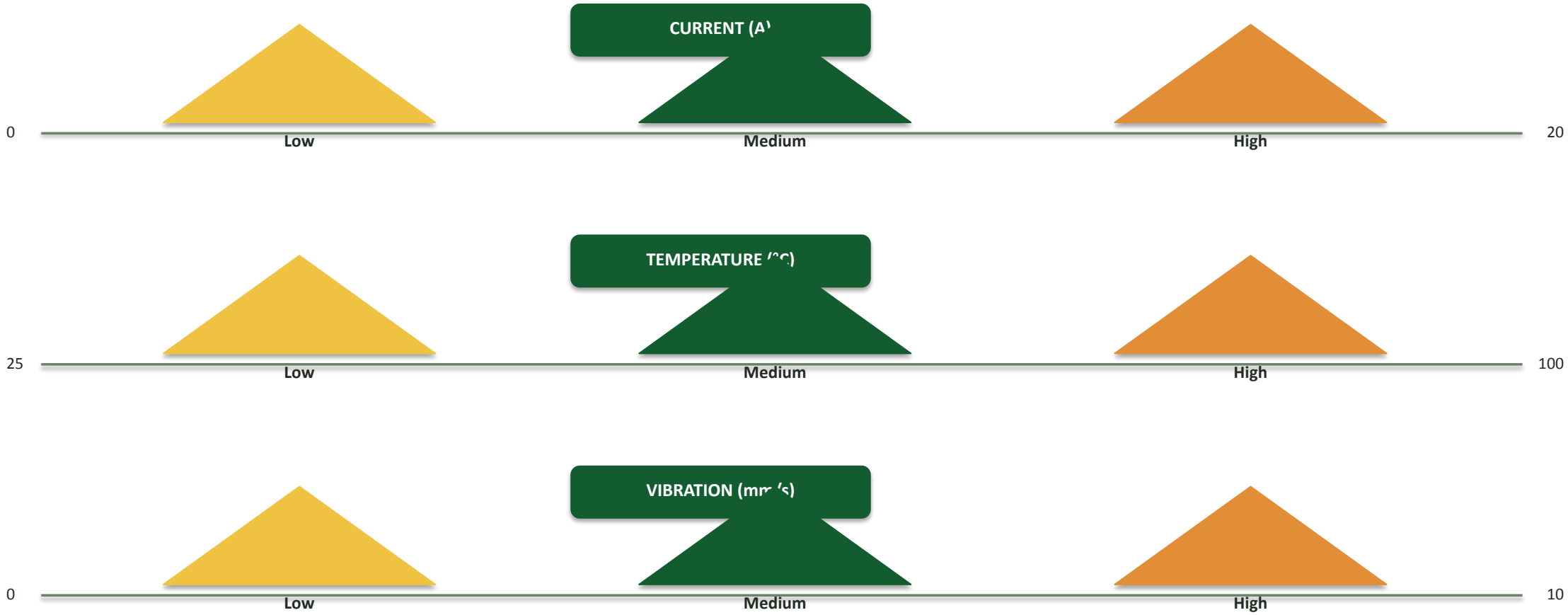
Current ⟶ electrical loading

Temperature ⟶ heating condition

Vibration ⟶ mechanical condition

Final output combines all three inputs to determine the motor-health status.

7. Membership Functions



Membership value lies between 0 and 1, allowing gradual decisions instead of rigid thresholds.

8. Fuzzy Rule Base (Example)

IF Current	AND Temperature	AND Vibration	THEN Condition
Low	Low	Low	Normal
Low	Medium	Low	Normal
Medium	Medium	Medium	Warning
High	Medium	Medium	Warning
High	High	Medium	Critical
High	High	High	Critical
Medium	High	High	Critical

Total Rules: 27 (3 × 3 × 3)

9. PBL Workflow (Phases)

Phase 1 Problem Identification – Identify motor failure monitoring problem.

Phase 2 System Design – Select parameters, sensors, fuzzy variables and rule base.

Phase 3 Implementation – Develop fuzzy system and Python simulation.

Phase 4 Testing – Test with different motor conditions.

Phase 5 Analysis – Analyze results and optimize rules using Genetic Programming.

Phase 6 UI Development – Create final interface and alerts.

10. Python Simulation & Genetic Programming

PYTHON SIMULATION

✓ Fuzzy logic implemented using a simple fuzzy library

✓ Take inputs from user

✓ Apply membership functions

✓ Apply fuzzy rules

✓ Defuzzification (Centroid method)

✓ Display condition and maintenance suggestion

GENETIC PROGRAMMING

• Chromosome = Rule set

• Fitness = Accuracy

• Selection, crossover and mutation applied

• Best rule set gives minimum error

Expected: optimized rules improve classification accuracy and reduce false alarms.

11. Expected Results & Analysis

EXPECTED RESULTS

✓ Accurate classification of motor condition

✓ Reduced false alarms

✓ Early fault detection

✓ Increased motor life

✓ Low maintenance cost

SAMPLE TEST

Current = 12.5 A
Temperature = 68 °C
Vibration = 4.2 mm/s

Motor Condition
WARNING

Suggestion:
Check the motor and schedule maintenance.

Overall Accuracy Improvement with GP Optimized Rules

12. Conclusion & Future Scope

FINAL TAKEAWAY

We convert three measurable motor conditions into one intelligent health decision using fuzzy logic, then use genetic programming to improve the decision rules.

Current + Temperature + Vibration

→ Fuzzy Logic →

Normal / Warning / Critical

FUTURE SCOPE

- Real sensors + microcontroller
- IoT / mobile alerts
- Predictive maintenance
- Energy and fault analytics

WHY THIS TOPIC

- Electrical branch relevant
- Easy to demonstrate
- Covers all PBL phases
- Strong presentation / viva topic

THANK YOU